SMARTBRIDGE
Let's Bridge the Gap

SKILLWALLET
A SmartBridge Product

Solution Requirements

Date	15 March 2026
Team ID	SWTID-2026-6775
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	**Authentication**	Users can securely register, log in, and verify identity using OAuth 2.0 (Google Sign-In) or JWT-based email authentication. Password reset and session management features are included.	**High**
2	**Authorization levels**	System defines distinct user roles: Standard Users (access personalized AI workouts, profile settings, and plan history) and Admin (manages system prompts, monitors API usage, and views platform analytics).	**High**
3	**External interfaces**	Integrates with Google Gemini API for context-aware AI fitness generation, cloud storage (MongoDB/Firestore) for profile persistence, and third-party health tracking services for data exchange.	**High**
4	**AI Fitness Generation**	Takes user inputs (age, weight, fitness goal, dietary restriction, available equipment) and queries Gemini models to construct structured, step-by-step weekly workout and diet plans.	**High**
5	**Plan History & Export**	Allows users to save generated fitness regimes to their account dashboard, track weekly adherence, and export plan details as a PDF or calendar file.	**Medium**

All PDF Reader

		services the system must connect to]	
4	Transactions processing	*[Explain how data exchanges, payments, or core system transactions will be handled]*	
5	Reporting	*[Describe the types of reports, dashboards, or data exports the system must generate]*	
6	Business rules, etc.	*[List specific logic, constraints, or workflows unique to the business process]*	
7	Compliance to laws or regulations	*[Specify any legal, industry, or data privacy standards the system must adhere to (e.g., GDPR, HIPAA)]*	
8	*Other*	*[Add any additional functional requirements here]*	

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	*[Describe response times, processing speed, etc.]*	*(e.g., Page loads in < 2 seconds)*
2	Scalability	*[Describe how the system handles growth in users or data]*	*(e.g., Supports up to 10,000 concurrent users)*
3	Security & Data Privacy	*[Detail data protection, encryption, and vulnerability management]*	*(e.g., End-to-end AES-256 encryption)*
4	Reliability & Availability	*[Specify uptime requirements and fault tolerance]*	*(e.g., 99.9% uptime per month)*
5	Usability & Accessibility	*[Describe user experience standards and accessibility compliance]*	*(e.g., Complies with WCAG 2.1 AA standards)*
6	*Other*	*[Add any additional non-functional requirements (e.g., Maintainability, Portability)]*	